Tatsuyuki Sekine

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Gojenbergsweg 112, 21029 Hamburg, Germany

EDUCATION

University of Hamburg Oct. 2025 – Present
Ph.D. in Physics Hamburg, Germany

- **Research**: Exploration of unsupervised machine learning to search for $z > 6$ quasars from the Euclid Wide Survey data
- **Advisors**: Dr. Jan-Torge Schindler, Dr. Francesco Guarneri

Tokyo Institute of Technology (Science Tokyo) Apr. 2017 – Mar. 2019
M.Sc. in Physics Tokyo, Japan

- **Thesis title**: Search for Z' boson resonance decaying into top quarks with the LHC-ATLAS experiment using machine learning
- **Advisors**: Prof. Masahiro Kuze, Dr. Yohei Yamaguchi

Tokyo Institute of Technology (Science Tokyo) Apr. 2013 – Mar. 2017
B.Sc. in Physics Tokyo, Japan

- **Thesis title**: Reconstructions of hadrons containing heavy quarks with the LHC-ATLAS experiment
- **Advisors**: Prof. Masahiro Kuze

Professional Experience

ELSPINA VEINZ INC. Aug. 2022 – Sep. 2025
Data science researcher Tokyo, Japan

- Conducted R&D of satellite remote sensing data analysis for real-time monitoring applications
- Consulted on the integration of generative AI into internal enterprise systems

Amanogi Corp. Aug. 2020 – Jul. 2022
Data science researcher Tokyo, Japan

- Development of a large-scale marine debris monitoring system using satellite remote sensing data and GIS

Softbank Corp. Apr. 2019 – Jul. 2020
Software engineer Tokyo, Japan

- Development of application software for the Internal system

Scholarship

DAAD Research Grants - Doctoral Programmes in Germany Oct. 2025 – Sep. 2028

SKILLS

Machine learning : Pytorch, Tensorflow, Keras, Schikit-learn
Programming : Python, Java, C++, JavaScript, HTML/CSS, SQL, Bash
MLOps : Git, Docker, AWS, GCP